HARSHITH VH

Bachelor of Science – Physics & Computer Science
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Education

Degree	Institution	Year	Score
BSc Physics & Computer Science	Kristu Jayanti (Deemed to be University)	2023–2026	8.7 / 10 (upto 5th sem)
XI – XII (Science)	The National Pre University College	2021–2023	87%

Internship

Research Internship – Cosmic Ray Laboratory, TIFR (NCRA), Ooty | June 2025

- Worked with the Muon Telescope setup at the Cosmic Ray Laboratory, Ooty, to study secondary cosmic ray muon flux at ground level.
- Operated and calibrated scintillator-based detectors used in the muon telescope array for particle detection and event triggering.
- Performed data acquisition, quality checks, and preliminary analysis of proportional counter data.
- Gained hands-on experience in high-energy particle detection instrumentation and data analysis pipelines in an experimental astrophysics environment.

Research Interests

Field	Topics
Gravitational Waves & Data Analysis	Compact Binary Objects, Q-transform Analysis, Wavelet Transforms, Time-Frequency Analysis, Bayesian Inference for Source Parameter Estimation
Machine Learning	Supervised and Unsupervised Algorithms, Deep Learning and Neural Networks
Exoplanet Science	Transit Curve Analysis, High-Contrast Imaging, PSF Subtraction (KLIP, PyKLIP), Flux Calibration & Planet Mass Estimation
Cosmic Ray Physics	Muon Detection, Scintillator-based Detectors, Data Acquisition & Calibration

Academic Coursework

- Classical Mechanics
- Quantum Mechanics
- Semiconductor Physics and Digital Electronics

- Condensed Matter Physics
- Atomic, Molecular and Nuclear Physics
- Mechanics and Properties of Matter
- Optics
- Heat and Thermodynamics
- Electricity and Magnetism
- Database Management System and SQL.
- Programming in C
- Data Structures and Algorithms
- Java Programming
- Internet Technology (HTML, CSS, JavaScript)
- PHP Programming
- Software Engineering
- Scientific Computing using Python – IIT Kanpur via NPTEL
- Applications of Modern Physics

Research & Projects

Q-Transform Analysis of GW230529 [Research] | 2024

- Performed Q-transform time-frequency analysis of GW230529, a suspected neutron star–black hole (NSBH) merger.
- Identified a significant signal between −9 and −9.2 s relative to trigger time, suggesting potential early inspiral or noise artefacts.
- Demonstrated the effectiveness of time-frequency methods in gravitational wave detection and interpretation.

Evolving Sensitivity of LIGO Livingston: Inspiral Range and Mass Scaling Across Observing Runs [Research] | 2025

- Analyzed LIGO Livingston detector sensitivity across O1, O2, and O3 using Amplitude Spectral Density (ASD) and noise characterization techniques.
- Computed inspiral ranges for BNS, BBH, and BHNS systems, demonstrating significant sensitivity improvement (e.g., BNS: ~63 Mpc → ~110 Mpc).
- Investigated chirp mass scaling relation, identifying deviation from theoretical $d \propto M^{(5/6)}$ with an observed power law $d \propto M^{0.67}$.
- Developed a Python-based data analysis pipeline using GWOSC data for ASD computation, interpolation, and numerical integration.
- Provided insights into detector noise reduction, mass-dependent sensitivity, and implications for future runs (O4).

Classification of Gravitational Wave Events Based on Sources | 2024

- Developing a machine learning model to classify GW events into BBH, BNS, and NSBH mergers using astrophysical parameters extracted from GWOSC data.
- Focused on improving event classification accuracy through feature engineering and model comparison.

Predicting Habitability of Exoplanets | 2024

- Applied Logistic Regression to classify exoplanet habitability using planet radius, mass, equilibrium temperature, stellar mass, and other features.
- Model predicts whether an exoplanet falls within the habitable zone.

Earthquake Magnitude Prediction Using Machine Learning | 2025

- Developed and optimised Random Forest, Gradient Boosting, and SVR models on historical Indian seismic data.
- Analysed model performance using statistical metrics and visualised prediction accuracy.

Rainfall Classification Using Machine Learning |2025

- Implemented Random Forest, SVM, and Logistic Regression to classify annual rainfall patterns in India into Low, Medium, and High categories.
- Applied feature engineering techniques on historical weather data to enhance model performance.

Gravitational Acceleration Using Linear Regression |2025

- Applied linear regression to free-fall experimental data to estimate gravitational acceleration.
- Modelled the relationship between fall-time squared and height using controlled drop measurements.

Ionic Plasma Thruster – Experimental Model | Kristu Jayanti College | 2025

- Developed a working prototype of an ionic plasma thruster that accelerates ionised particles via a high-voltage electric field.
- Explored feasibility for satellite propulsion and deep-space mission applications, comparing efficiency against conventional chemical rockets.

Campus Sphere – Full-Stack Web Application | 2025

- Building a full-stack campus social and resource platform featuring event discovery, club management, academic resource sharing, and student networking.
- Designed a scalable backend architecture with a responsive UI, focusing on seamless user experience for college communities.

Presentations

1. *Harshith V. H.*, Vaishnavi K., B. Hariharan (Tata Institute of Fundamental Research), et al.
   "Characteristic Analysis of Proportional Counters and Plastic Scintillators Used in the GRAPES-3 Experiment"
   National Conference [NCARAS-2025], *MVJ College of Engineering*, Bengaluru
   (Focus: detector characterization, cosmic ray instrumentation, GRAPES-3 experiment)

Workshops [Conducted]

Title	Event / Venue	Year
Introduction to Gravitational Waves and Data Analysis	Student Enrichment Program for BSc Physics Students	2026
Exoplanet: Transit Curve Analysis and Advanced Imaging	Student Enrichment Program for BSc Physics Students	2024
Hands-On Session on Transit Curve Analysis	Student Enrichment Program for BSc Physics Students	2024
Cosmic Conundrums – Host	National Space Day	2024

Workshops & Schools [Attended]

Title	Institution	Year
ISRO START Program 2026	Indian Space Research Organisation (ISRO)	2026
LIGO Undergraduate Study Hub	California Institute of Technology	2025
PCB Designing Workshop	Altium	2025
ICTS-TIFR Gravitational Waves Open Data Workshop	ICTS, Bangalore	2024
NASA Sagan Workshop	California Institute of Technology	2024
Pravega Astrophysics Workshop	IISc, Bangalore	2023

Awards & Honours

Award	Institution / Event	Year
First Prize – Poster Presentation, National Conference NCARAS-2025	MVJ College of Engineering (Autonomous), Bengaluru	2025
Synergia – Physics Fest Winner	Kristu Jayanti College	2024
Short Film Making – Winner	Kristu Jayanti College	2023
Merit – XII	The National Pre University College	2023

Technical Skills

- **Programming:** Python, C, C++, Julia, Java
- **Machine Learning:** Scikit-Learn, TensorFlow, XGBoost, Random Forest, SVM, PCA
- **Gravitational Wave Analysis:** GWpy, PyCBC, Bilby, Q-transform, Bayesian Inference
- **Exoplanet Science:** PyKLIP, PSF Subtraction, Flux Calibration, Transit Curve Analysis
- **Web Development:** HTML, CSS, JavaScript, PHP, MySQL, React
- **Tools & Software:** LaTeX, Jupyter Notebook, Git, Linux (Ubuntu)
- **Data Analysis & Visualisation:** Pandas, NumPy, Matplotlib, Seaborn

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